

Streaming Subject Linking: Similarity Threshold Baseline

Step 5 — Task 2a: how far does cosine similarity alone get us?

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1. Objective

Task 2 of the Knesset Protocol Journal Project is *streaming subject linking*: as protocol spans arrive chronologically, classify each as **LINK** (belongs to an existing journal entry) or **NEW** (opens a fresh entry).

This step implements and evaluates the simplest possible baseline: a **cosine similarity threshold** over the best representations from Steps 3–4. No LLM is involved. The goals are:

1. Establish a quantitative floor for linking performance.
2. Characterize the precision–recall trade-off, especially the **false-merge rate** (FMR), which is asymmetrically costly: a false merge irreversibly conflates two distinct legislative matters in the journal.
3. Determine whether the baseline is sufficient or whether an LLM verifier is needed.

2. Method

2.1 Task setup

Each of the 198 clean spans has a gold canonical topic label (Step 1). 28 topics recur across ≥ 2 protocols (105 spans); 93 topics appear exactly once.

Gold labels for the streaming decision:

- A span’s *first* chronological occurrence $\rightarrow$ gold = **NEW**.
- All subsequent occurrences of the same topic $\rightarrow$ gold = **LINK**.

This yields 77 LINK instances and 121 NEW instances across the 198 spans.

2.2 Decision rule

Let $R = \{r_1, \dots, r_m\}$ be the current set of journal-entry representatives (one embedding per entry, initialized to the first span seen for that entry). For an incoming span s :

$$\hat{y}(s; \theta) = \begin{cases} \text{LINK to } \arg \max_j \cos(s, r_j) & \text{if } \max_j \cos(s, r_j) \geq \theta, \\ \text{NEW} & \text{otherwise.} \end{cases}$$

We sweep $\theta \in [0, 1]$ in steps of 0.005 to obtain full precision–recall and false-merge-rate curves.

2.3 Evaluation modes

Pairwise (oracle upper bound). For all $\binom{105}{2}$ ordered pairs within the recurring subset, predict LINK if $\cos(i, j) \geq \theta$. This assumes perfect access to all past spans and is the ceiling on what the similarity function can achieve.

Streaming simulation (realistic). Process all 198 spans in chronological order. Journal entries are represented by the first span seen for each cluster (no centroid update). Errors propagate: a wrong link joins two clusters permanently.

2.4 Representations tested

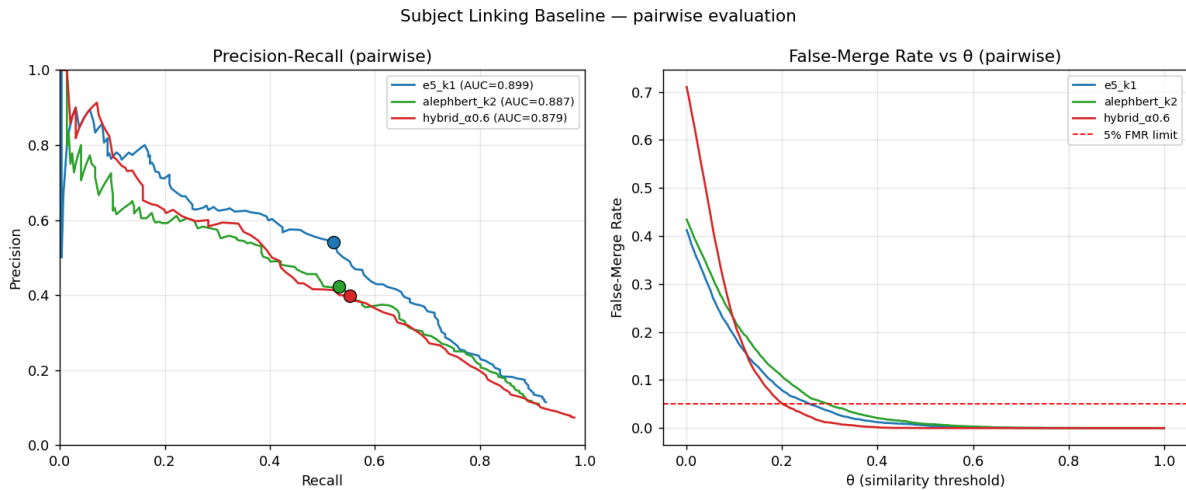
Name	Construction	Why
e5_k1	ABTT $k = 1$ on e5-large	Best AUC (0.899) from Step 3
alephbert_k2	ABTT $k = 2$ on AlephBERT	Best ARI_{agg} from Step 3
hybrid $_{\alpha=0.6}$	$0.6 \cdot S_{\text{tfidf}} + 0.4 \cdot S_{\text{alephbert_k2}}$	Best precision-priority from Step 4

3. Results

3.1 Pairwise evaluation

Representation	AUC	Best F1	θ	F1 ($\text{FMR} \leq 5\%$)
e5_k1	0.899	0.531	0.33	0.503
alephbert_k2	0.887	0.472	0.32	0.464
hybrid $_{\alpha=0.6}$	0.879	0.463	0.20	0.463

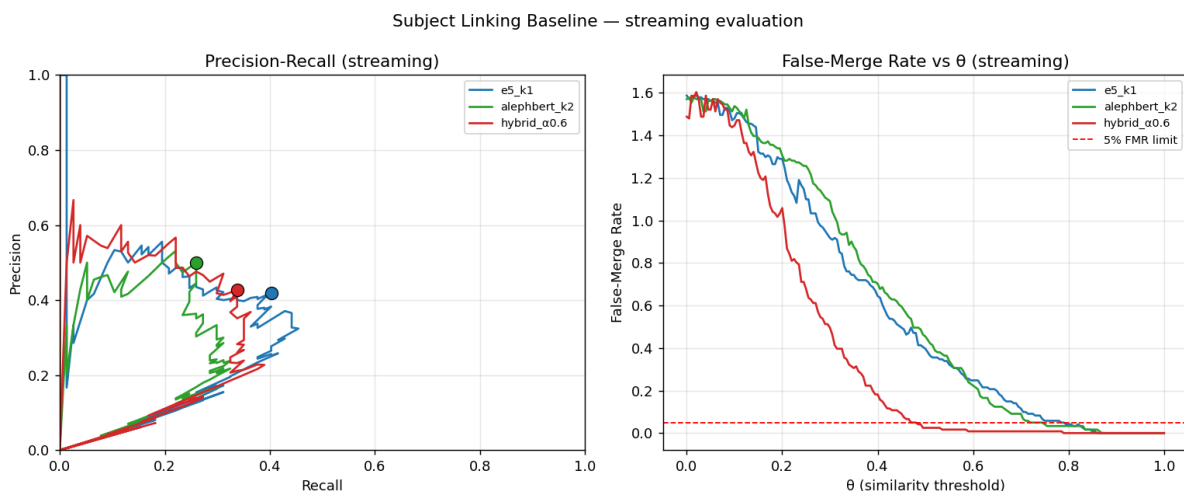
e5_k1 leads on all metrics. At the precision-priority operating point ($\text{FMR} \leq 5\%$) the pairwise F1 drops only slightly ($0.531 \rightarrow 0.503$ at $\theta = 0.26$), suggesting the score distributions are reasonably well-separated in the oracle setting.



Left: precision–recall curves (filled circles = F1-optimal threshold per model). Right: false-merge rate vs. θ (dashed line = 5% FMR budget).

3.2 Streaming simulation

Representation	Best F1	θ	P	R	F1 ($\text{FMR} \leq 5\%$)
e5_k1	0.411	0.52	0.419	0.403	0.092
alephbert_k2	0.342	0.62	0.500	0.260	0.114
hybrid $_{\alpha=0.6}$	0.377	0.36	0.426	0.338	0.196



Same layout as above but for the streaming simulation.

3.3 Analysis of the pairwise-streaming gap

The best pairwise F1 (0.531) drops to 0.411 in the streaming simulation for **e5_k1** — a gap of 0.12. This is caused by the **first-span-representative** approximation: the first occurrence of a topic is often an introductory or procedural discussion whose embedding is not representative of the full legislative matter. Later spans, which discuss the matter in depth, are compared against this impoverished representative.

3.4 The anti-duplication operating point

At the precision-priority constraint ($\text{FMR} \leq 5\%$), streaming recall collapses:

- **e5_k1**: $\text{F1} = 0.092$ at $\theta = 0.79$ — only the most obvious recurring spans are linked.
- **hybrid $_{\alpha=0.6}$** : $\text{F1} = 0.196$ at $\theta = 0.47$ — TF-IDF's bill-number precision prevents false merges at a lower threshold, recovering more recall.

This is the central finding: **no single threshold satisfies both the anti-duplication constraint ($\text{FMR} \leq 5\%$) and useful recall ($\text{F1} \geq 0.3$)** using similarity alone. The distributions of same-topic and different-topic cosine similarities overlap too much in the streaming setting for a threshold to cleanly separate them.

4. Takeaways

- **Pairwise AUC is strong** (0.899) but does not translate to high streaming F1. The bottleneck is the first-span-representative, not the similarity function itself.
- **The similarity baseline is insufficient for anti-duplication deployment.** At $\text{FMR} \leq 5\%$, the best streaming F1 is 0.196. A journal system using only similarity thresholds would either miss most recurrences or create false merges.
- **Hybrid $\alpha = 0.6$ is the best precision-priority choice** because TF-IDF bill-identifier signal is discriminative at the precision-critical margin.
- The gap between pairwise and streaming performance points directly to **centroid-based entry representation** as the quickest engineering fix, and to **LLM-based verification** as the path to production-quality linking.

5. Recommended Operating Points

Use case	Representation	θ
Recall-maximizing (false merges acceptable)	e5_k1	0.33
Precision-priority journal ($\text{FMR} \leq 5\%$)	hybrid $_{\alpha=0.6}$	0.47
Retrieval pre-filter for LLM verifier	e5_k1	rank, don't threshold

6. Future Directions

1. **LLM retrieve-then-verify (Task 2b).** Use e5_k1 to retrieve the top- K journal candidates ($K = 3\text{--}5$, $\theta = 0.33$) and prompt a Hebrew-capable LLM to classify each as *same* / *related* / *new* given the candidate's full prior timeline. This is the critical next step.
2. **Centroid-based entry representation.** Replace the first-span representative with a running centroid over all ABTT-corrected embeddings linked to the entry. Expect to close a significant portion of the pairwise-streaming gap.
3. **Annotated evaluation set.** Manually label a stratified sample of span pairs for LINK / RELATED / NEW to enable three-class evaluation and remove reliance on topic-label proxy supervision.
4. **Asymmetric threshold tuning.** Calibrate separately on topics known to be legislatively similar (budget sub-items, closely related bills) vs. unrelated topics to narrow the hard overlap region.

Artifacts (this step): outputs/linking_results.json, outputs/pr_curves_pairwise.png, outputs/pr_curves_st

Code: code/linking_baseline.py.